

# Water Treatment & pH Quick Reference

The 5-step Cochise bicarbonate protocol · distilled from Guide 13 §5 + Guide 4 Part 3

## The 5-Step Cochise Bicarbonate Protocol

\*Guide 13 §5\*

Run every time you fill a hydroponic reservoir, and whenever pre-mixing raw well water for sensitive crops. Chemistry in Guide 13 §6.

| # | Action                   | Detail                                                                                                                                                                     | What happens                                                                                                        |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1 | Fill                     | Source water to working volume (≈12 gal in a 15-gal tub). Let stand 15 min if cold from well. Record starting pH, EC, alkalinity.                                          | Starting pH 7.5–8.5; EC 0.3–0.8 mS/cm; alkalinity 150–300+ ppm CaCO <sub>3</sub> for most Cochise wells.            |
| 2 | Acid in                  | Add acid slowly while stirring. Citric stock (1 g/mL), commercial phosphoric pH Down, or distilled vinegar. Dose 0.5–1 mL/gal to start. Target pH <b>5.0</b> in this step. | Bicarbonates react with acid — water visibly fizzes as CO <sub>2</sub> releases. The fizz is the chemistry working. |
| 3 | Wait for fizzing to stop | Stir gently every minute. Fizz dies off as bicarbonates run out. Typically 5–15 min. Do not move on until the surface is calm.                                             | All bicarbonates have converted to CO <sub>2</sub> and left the reservoir. The buffer is now gone.                  |
| 4 | Measure final EC         | Read EC after fizzing stops. Should match starting EC + small acid contribution (+0.05–0.15 mS/cm).                                                                        | If EC dropped meaningfully, alkalinity reading was wrong; if it jumped >0.2 mS/cm, you overshot acid.               |
| 5 | Adjust to target pH      | Small additional doses of pH Down (or pH Up if you overshot). Wait 5 min between checks.                                                                                   | Final water is at target pH, stable, ready for nutrients (hydro) or direct use (irrigation).                        |

### WARNING

**Never skip Step 3.** Adding nutrients while bicarbonates are still active causes Ca/Mg to precipitate as carbonate scale within hours — cloudiness, white film, chalky residue. Nutrients lock up; plants starve. Single most common failure mode in Dragoons hydroponics. Wait for the calm surface. **Always add acid to water, never water to acid** — pouring water onto concentrated acid can cause sudden boiling. Safety glasses + gloves for anything stronger than vinegar.

## pH Targets by Use

\*Guide 13 §5.4\*

| Use                                       | Target pH        | Notes                                       |
|-------------------------------------------|------------------|---------------------------------------------|
| Hydroponic peppers, tomatoes, cucurbits   | <b>5.8 – 6.2</b> | Standard fruiting-crop range.               |
| Hydroponic leafy greens (lettuce, herbs)  | 5.8 – 6.3        | Slightly broader tolerance.                 |
| Hydroponic brassicas                      | 6.0 – 6.5        | Brassicas prefer slightly higher.           |
| Soil-bed drip irrigation, most vegetables | <b>6.5 – 7.0</b> | Headroom for biological drift.              |
| Orchard / mature trees                    | <b>6.0 – 7.5</b> | Trees are tolerant; minimize acid.          |
| Acid-loving crops (blueberry, azalea)     | 5.0 – 5.5        | Difficult crops in the Dragoons regardless. |

## Choosing the Acid

\*Guide 13 §5.3 + Guide 4 Part 3\*

| Acid                                        | Strength    | Notes                                                                                                                                       |
|---------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Citric acid</b> (powder, 1 g/mL stock)   | Precise     | <b>Best purchased option.</b> Accurate dosing, no surprises. Standard choice for the protocol. ~\$10/lb at any grocery or canning supplier. |
| <b>Phosphoric acid</b> (commercial pH Down) | Very strong | <b>Most reliable for hard well water.</b> Adds a small amount of phosphorus — a benefit at fruiting, not a problem.                         |
| <b>Prickly-pear vinegar</b> (for-aged)      | pH 3–4      | <b>Best local option</b> once you have a patch. Ferment crushed prickly-pear juice open to air 2–4 weeks. Gentle, easy to dose.             |
| <b>Distilled white vinegar</b> (5% acetic)  | Mild        | Workable for small batches — but consumes a lot of volume (4–8 oz per 15-gal tub). Household backup, not routine.                           |
| <b>Sulfur burn</b>                          | pH 1–2      | Strong; fumes are irritating; outdoors only. Not recommended for routine use.                                                               |

### LOCAL TIP

**Source quality ranked** (cleaner = less protocol work): monsoon rainwater ●●●● > RO well ●●●● > 50:50 RO/rain blend ●●●● > treated well ●●●● > untreated well ●●●● (do not skip protocol). **The Dragoons monsoon collapses the bicarbonate problem mid-July through October** — rainwater catchment reduces Step 2 to a small pH nudge. High-leverage capture window is July.